

1. The Photography Trap

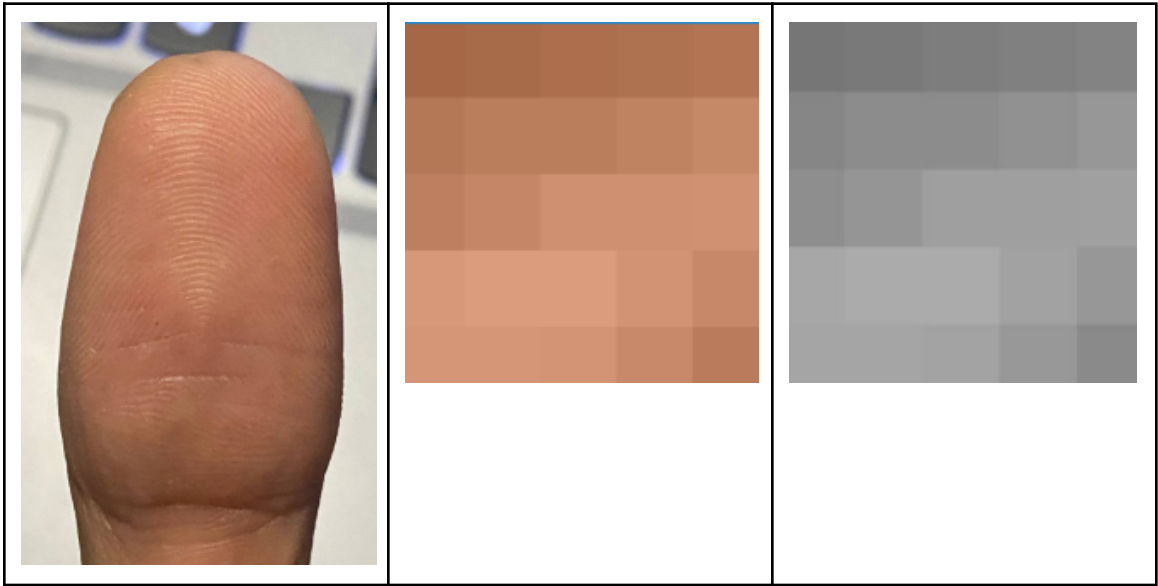
The white surface in the dark shadow has a lower pixel intensity than the black surface in bright sunlight because the black surface in bright sunlight is illuminated much greater. Our brain identifies the shadowed object as white because of the color constancy. We humans do not really solely focus on the illumination on the subject, but we consider things such as the surrounding context. I.e: We know that the white surface is inside a dark place so we cannot automatically say that it's black.

2. The Macro Mystery



I chose a leather wallet. When I took a macro picture of the leather and showed it to my friend, they couldn't tell if it was a shoe, a belt, a bag, etc. They needed the overall appearance of the wallet to fully see that the photo was a picture of a wallet as they needed to see the features of the wallet such as the way it folds, what's inside, the stitches, the size, the shape, etc. This shows that object recognition depends not only on the local features but also on the global context and relationship between the different parts of the object.

3. The Matrix Reality



The fingerprint disappeared and became a data structure when I represented the colors of the image as a matrix instead of looking at the actual visual outlook of the colors. I basically converted the image to something a computer can process and this shows how the computer vision system can represent real-world objects as numbers and mathematical structures they can operate on instead of experiencing it the way a human does.

4. The Feature Trace



Yes the objects became a lot more distinguishable when I outlined them, I can clearly see the shape, though when I highlighted the regions, it became quite hard to tell what some objects were.